A Power Model for D-MIMO Systems: Comparing the Energy Consumption of D-MIMO and Co-located MIMO

Thomas Feys, François Rottenberg
KU Leuven, Dept. ESAT-WaveCore, Dramco, B-9000 Ghent, Belgium

***Abstract*—TODO**
***Index Terms*—**

I. Introduction

TODO small paragraph on energy consumption of 5G 6G networks

Recent works have focused on characterizing this energy consumption of wireless networks, and in particular of multiple input multiple output (MIMO) systems. In [1], the authors propose a detailed power consumption model for collocated MIMO systems, which is based on the power consumption of the different hardware components in the system for frequency range 3 (FR3) which is envisioned for 6G systems. Another prevelant enabler for 6G systems is distributed MIMO (D-MIMO)/cell-free MIMO. However, works that characterize the power consumption of D-MIMO systems is limited. TODO add som refs and work here Moreover, it is unclear whether the power consumption of D-MIMO systems is is on par, lower or higher than the one of collocated MIMO systems. Especially, considering the additional network consumption that occurs from having multiple access point (AP), fronthaul links, synchronization, etc. In this work we introduce an extension of the power model introduced in [1] for the D-MIMO scenario. We incorporate TODO mention all extensions. TODO mention it is opensource an where it can be found.

II. D-MIMO System Model

We consider a time division duplexing (TDD) D-MIMO network in which L APs, each with M antennas, jointly serve K single-antenna user equipments (UEs) over Q orthogonal frequency division multiplexing (OFDM) subcarriers, resulting in LM total antennas. All L APs are connected to a central processing unit (CPU) by fronthaul links. The indices $l = 1, \ldots, L$, $m = 1, \ldots, M$, $k = 1, \ldots, K$ and $q = 1, \ldots, Q$ denote the number of APs, antennas per AP, UEs and subcarriers respectively. Co-located massive MIMO (mMIMO) is the special case $L = 1$ with $M_{\text{ant}} = LM$ antennas on a single site at the centre of the coverage area. Co-located and distributed MIMO consider the same number of total antennas and radiate the same total power, they only differ in where the antennas sit.

We consider the same frame structure as in [1]. Namely, the total frame length is T_f which is split into a downlink (DL) and uplink (UL) phase of length $\tau_{\text{DL}} T_f$ and $\tau_{\text{UL}} T_f$ respectively, with $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$. We consider N_{DL} available OFDM symbols in DL and N_{UL} in UL, of which $N_{\text{a,DL}}$ and $N_{\text{a,UL}}$ are activated and carrying data. The activated fraction is the average physical resource load

$$\bar{x}_i = \frac{N_{\text{a},i}}{N_i} \in [0, 1] \quad i \in \{\text{DL}, \text{UL}\}, \tag{1}$$

so $\bar{x}_i = 1$ is a fully loaded direction and $\bar{x}_i = 0$ one that carries reference signals only. In general, as defined in [1], the load also counts how many UEs are served and how many subcarriers each of them is allocated. We assume throughout that every activated symbol serves all K UEs on all Q subcarriers, so the load reduces to the ratio of activated to available symbols. The UL and DL frames are further subdevided into three parts as depicted in TODO add fig and ref it. The first part is for reference signaling and has length

$$\tau_i \tau_{\text{sig},i} \quad i \in \{\text{DL}, \text{UL}\}, \tag{2}$$

where $\tau_{\text{sig},i} \in [0, 1]$ is the fraction of the frame used for signaling. The second part is for data transmission and has length

$$\tau_i (1 - \tau_{\text{sig},i}) \bar{x}_i \quad i \in \{\text{DL}, \text{UL}\}. \tag{3}$$

The third part represents a micro-sleep or discontinuous transmission (DTX) mode [2] and has length

$$\tau_i (1 - \tau_{\text{sig},i})(1 - \bar{x}_i) \quad i \in \{\text{DL}, \text{UL}\}. \tag{4}$$

A. Signal Model

Let $\mathbf{h}_{kl}[q] \in \mathbb{C}^M$ be the channel from AP l to UE k. Stacking the APs gives the collective channel $\mathbf{h}_k[q] = [\mathbf{h}_{k1}[q]^\mathsf{T} \ \cdots \ \mathbf{h}_{kL}[q]^\mathsf{T}]^\mathsf{T} \in \mathbb{C}^{LM}$ and $\mathbf{H}[q] = [\mathbf{h}_1[q] \ \cdots \ \mathbf{h}_K[q]] \in \mathbb{C}^{LM \times K}$.

1) Downlink: AP l transmits $\mathbf{x}_l[q] = \sum_k \mathbf{w}_{kl}[q] s_k[q]$, with $\mathbf{w}_{kl}[q] \in \mathbb{C}^M$ its precoding vector for UE k and $s_k[q]$ the unit-power data symbols, mutually independent across UEs. Let $\mathbf{w}_k[q] \in \mathbb{C}^{LM}$ be the collection of the L precoders of the APs $\mathbf{w}_{kl}[q]$, the signal received by UE k is

$$y_k[q] = \mathbf{h}_k[q]^H \mathbf{w}_k[q] s_k[q] + \sum_{i \neq k} \mathbf{h}_k[q]^H \mathbf{w}_i[q] s_i[q] + n_k[q] \tag{5}$$

thanks

with $n_k[q] \sim \mathcal{CN}\left(0, \sigma^2/Q\right)$. Treating the multi-user interference as noise,

$$\mathrm{SINR}_k^{\mathrm{DL}}[q] = \frac{\left|\mathbf{h}_k[q]^H \mathbf{w}_k[q]\right|^2}{\sum_{i \neq k} \left|\mathbf{h}_k[q]^H \mathbf{w}_i[q]\right|^2 + \sigma^2/Q}, \tag{6}$$

where σ^2 is the receiver noise power.

2) Uplink: UE k transmits with power $p_k[q] = p_k/Q$ subject to its own budget $p_k \leq p_{\max}$, and AP l observes

$$\mathbf{y}_l[q] = \sum_k \sqrt{p_k[q]}\, \mathbf{h}_{kl}[q] s_k[q] + \mathbf{n}_l[q] \tag{7}$$

with $\mathbf{n}_l[q] \sim \mathcal{CN}\left(\mathbf{0}, (\sigma^2/Q)\mathbf{I}_M\right)$. Applying a receive combiner $\mathbf{v}_k[q] \in \mathbb{C}^{LM}$ gives

$$\hat{s}_k[q] = \mathbf{v}_k[q]^H \mathbf{y}[q] = \sum_{l=1}^{L} \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]. \tag{8}$$

The resulting signal-to-interference-plus-noise ratio (SINR) is

$$\mathrm{SINR}_k^{\mathrm{UL}}[q] = \frac{p_k \left|\mathbf{v}_k[q]^H \mathbf{h}_k[q]\right|^2}{\sum_{i \neq k} p_i \left|\mathbf{v}_k[q]^H \mathbf{h}_i[q]\right|^2 + \sigma^2 \left\|\mathbf{v}_k[q]\right\|^2}. \tag{9}$$

B. Channel and Noise Model

Channels are drawn from the 3GPP TR 38.901 urban microcell model [3] through the Sionna implementation [4]. Each AP is a 38.901 base station carrying an M-element half-wavelength uniform linear array (ULA) of omnidirectional elements at height h_{AP}, and each UE is a single-antenna outdoor terminal at height h_{UE}. UEs are dropped uniformly over a square coverage area of side D, as are the APs. In the co-located case a single base station (BS) is placed at the centre. The receiver noise power follows from the thermal noise density, $\sigma^2\,[\mathrm{dBm}] = -174 + 10\log_{10}(B/\mathrm{Hz}) + F$, with B the bandwidth and F the noise figure in dB.

The model is specified through realizations rather than a closed-form correlation matrix, so the large-scale fading coefficient β_{kl}, used for power control, is estimated from the realization itself as

$$\beta_{kl} = \frac{1}{MQ} \sum_{q=1}^{Q} \left\|\mathbf{h}_{kl}[q]\right\|^2. \tag{10}$$

C. Centralized and Distributed Operation

Unlike the co-located case, which has a single sum-power budget, each AP has its own power constraint. Since the symbols are independent and unit-power, the power radiated by AP l over the OFDM block is

$$P_{T,l} = \sum_{q=1}^{Q} \sum_{k=1}^{K} \left\|\mathbf{w}_{kl}[q]\right\|^2 \leq \rho_{\max}. \tag{11}$$

How (11) is met is the difference between centralized and distributed operation, which is also related to the functional splits of Section III-A1.

1) Centralized operation (splits S1, S2): The CPU holds $\mathbf{H}[q]$ and designs all precoders jointly. In the downlink it applies zero forcing (ZF),

$$\bar{\mathbf{W}}[q] = \mathbf{H}[q]\left(\mathbf{H}[q]^H \mathbf{H}[q]\right)^{-1} \in \mathbb{C}^{LM \times K}, \tag{12}$$

which nulls all inter-user interference and requires $K \leq LM$. A single power coefficient ρ_k is attached to each UE, since the precoding direction already fixes how that power spreads over the LM antennas. Fractional power control weights the UEs by their aggregate gain $\beta_k = \sum_l \beta_{kl}$,

$$\rho_k = P_{\mathrm{tot}} \frac{\beta_k^v}{\sum_i \beta_i^v}, \qquad P_{\mathrm{tot}} = L\rho_{\max}, \tag{13}$$

with the exponent v trading throughput against fairness[1]. This power normalization is applied as

$$\mathbf{w}_k[q] = \gamma\sqrt{\rho_k}\, \frac{\bar{\mathbf{w}}_k[q]}{\sqrt{\sum_{q'} \left\|\bar{\mathbf{w}}_k[q']\right\|^2}}, \qquad \gamma = \sqrt{\frac{\rho_{\max}}{\max_l \tilde{p}_l}}, \tag{14}$$

where $\tilde{p}_l$ is the power AP l would radiate before that rescaling. Only the busiest AP then reaches its budget. Since $\sum_k \rho_k = L\rho_{\max}$ the nominal per-AP powers average to $\rho_{\max}$, so $\gamma \leq 1$, the constraint is always met. The uplink counterpart forwards the raw observations to the CPU, which combines over all LM antennas with $\mathbf{V}[q] = \mathbf{H}[q](\mathbf{H}[q]^H\mathbf{H}[q])^{-1}$.

2) Distributed operation (split S3): Each AP designs its own precoder/combiner from its local channel state information (CSI), so it suppresses only the interference it generates or observes itself and the coherent cross-AP cancellation is lost. Writing $\mathbf{E}_l[q] = [\,\mathbf{h}_{1l}[q] \ \cdots \ \mathbf{h}_{Kl}[q]\,] \in \mathbb{C}^{M \times K}$ for the local channel matrix and $\mathbf{P} = \mathrm{diag}\,(p_1, \ldots, p_K)$,

$$\mathbf{W}_l[q] = \left(\mathbf{E}_l[q]\mathbf{E}_l[q]^H + \lambda \mathbf{I}_M\right)^{-1} \mathbf{E}_l[q], \tag{15}$$

$$\mathbf{V}_l[q] = \left(\mathbf{E}_l[q]\mathbf{P}\mathbf{E}_l[q]^H + \lambda \mathbf{I}_M\right)^{-1} \mathbf{E}_l[q], \tag{16}$$

are the local regularized zero forcing (RZF) precoder/combiner which correspond to and the local minimum mean squared error (MMSE) precoder/combiner, with $\lambda = \sigma^2$. Each AP also allocates its own power budget among the UEs from local statistics,

$$\rho_{kl} = \rho_{\max} \frac{\beta_{kl}^v}{\sum_i \beta_{il}^v}, \tag{17}$$

so $\sum_k \rho_{kl} = \rho_{\max}$ by construction and every AP meets (11) with equality, $P_{T,l} = \rho_{\max}$, without any CPU-side rescaling. In the uplink the CPU fuses the L local estimates by summing them, $\hat{s}_k[q] = \sum_l \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]$.

In the uplink fractional power control is the same for both centralized and distributed operation at $p_k = p_{\max}\beta_k^{v_{\mathrm{ul}}} / \max_i \beta_i^{v_{\mathrm{ul}}}$. This is normalized by a maximum rather than a sum because each UE has its own independent power budget.

[1] A positive v follows the channel and serves the UEs with the largest aggregate gain first, which maximizes the sum rate but starves those at the edge of the coverage area. A negative v compensates the weak UEs: at $v = -1$ the product $\rho_k\beta_k$ is the same for every UE, so all of them are received at approximately equal signal power, which is the fair extreme. Values in $[-1, 1]$ therefore interpolate between a throughput-optimal and a max-min-fair allocation.

D. Delivered Rates

The per-UE spectral efficiencies average over the subcarriers and over the Monte Carlo channel realizations,

$$\mathrm{SE}_k^i = \tau_i(1-\tau_{\mathrm{sig},i})\,\bar{x}_i\,\mathbb{E}\left(\frac{1}{Q}\sum_{q=1}^{Q}\log_2\!\left(1+\mathrm{SINR}_k^i[q]\right)\right) \tag{18}$$

with $i\in\{\mathrm{DL},\mathrm{UL}\}$, the prelog is the data fraction (3) of the frame, since reference signalling and the deactivated symbols carry no payload. The delivered sum rates can then be written as $R_i = \tilde{B}\sum_k \mathrm{SE}_k^i$, with the effective bandwidth $\tilde{B} = 0.9\,B$.

III. Power Model Description

We first describe the full power consumption of a distributed network (Section III-A), then we focus on the two contributions that are specific for D-MIMO, i.e., the fronthaul and the synchronization of physically separate nodes (Section III-B). Finally we describe the parametric model of the AP consumption, which is based on [1] (Section III-C).

The topology, the indices l, m, k, q, the per-AP budget $\rho_{\max}$ and radiated power $P_{T,l}$, the frame structure and the delivered rates $R_{\mathrm{DL}}, R_{\mathrm{UL}}$ are those of Section II, and $i\in\{\mathrm{DL},\mathrm{UL}\}$ indexes the link direction. The APs are fully digital, so each carries M radio frequency (RF) chains. Every AP is assumed active and serves every user, and both directions are evaluated at full physical resource load, $\bar{x}_{\mathrm{DL}} = \bar{x}_{\mathrm{UL}} = 1$, so all APs share one network-wide operating load.

A. Network Power Consumption

The network consumption collects the L APs, the L fronthaul links that tie them to the CPU, and the CPU itself,

$$P_{\mathrm{net}} = \sum_{l=1}^{L}\left(P_{\mathrm{AP},l} + \frac{\overline{P_{\mathrm{FH},l}}}{\eta_{\mathrm{FH,sc}}}\right) + \frac{\overline{P_{\mathrm{CPU}}}}{\eta_{\mathrm{CPU,sc}}}, \tag{19}$$

where an overbar denotes a frame-averaged quantity and $\eta_{\mathrm{FH,sc}}, \eta_{\mathrm{CPU,sc}} \in (0,1]$ are the supply-and-cooling efficiencies of the fronthaul transceivers and of the central unit. The latter acts as an inverse power-usage effectiveness: the CPU sits in a cooled room rather than in an outdoor AP enclosure, so it should not be assigned the values fitted for a macro BS. Each active AP retains the three-block structure of the co-located model from [1] and adds the synchronization of Section III-B2,

$$P_{\mathrm{AP},l} = \frac{\overline{P_{\mathrm{dig},l}}}{\eta_{\mathrm{dig,sc}}} + \frac{\overline{P_{\mathrm{ana},l}}}{\eta_{\mathrm{ana,sc}}} + \frac{\overline{P_{\mathrm{PA},l}}}{\eta_{\mathrm{PA,sc}}} + \frac{P_{\mathrm{sync}}}{\eta_{\mathrm{sync,sc}}}, \tag{20}$$

with the digital, analog and amplifier blocks detailed in Section III-C. The co-located baseline is recovered by setting $L = 1$, $M = M_{\mathrm{ant}}$ and dropping the synchronization, fronthaul and CPU terms.

TABLE I: Node at which each digital operation runs. The uplink combiner follows the downlink precoder. Channel estimation is listed for completeness but is not priced, since perfect CSI is assumed.

Operation	S1	S2	S3
Encoding / decoding	CPU	CPU	CPU
Channel estimation	CPU	CPU	AP
Precoder computation	CPU	CPU	AP
Precoder application	CPU	AP	AP
OFDM, predistortion, BB filter	AP	AP	AP

1) Functional split: A single co-located site runs every digital operation in one place. A distributed network does not, and where each operation runs sets both the digital consumption and the fronthaul load. We consider the three splits of Table I. Split S1 forwards precoded samples: the CPU encodes, computes the centralized precoder and applies it, so AP l receives the M streams of frequency-domain samples. Split S2 forwards precoder coefficients and data: the CPU encodes and computes the precoder, but AP l applies it locally. Split S3 is fully local operation: the CPU only encodes, and each AP computes and applies its own precoder from local CSI. Splits S1 and S2 realize the same centralized precoder and therefore deliver the same rate; they differ only in where the work runs and in what crosses the fronthaul. S3 delivers a lower rate due to local processing.

2) MIMO processing and its placement: TODO the computational complexities will depend on the precoer type so move this to a table and reference it here. Following [1], a digital block consumes a static part plus its complex operations per sample Ξ evaluated at the constellation sampling rate $f_{s,\mathrm{I}}$ and divided by the computational efficiency η of the implementation. Two MIMO operations must be kept apart, because the split moves them independently. Applying an already computed precoder to K users over N chains costs $2KN$ operations per sample. Computing it costs the Cholesky terms of [1], amortised over the υ_{coh} samples of a coherence block. Under centralized operation the CPU inverts the $K\times K$ Gram matrix of the collective channel $\mathbf{H}[q]\in\mathbb{C}^{LM\times K}$, which is the co-located count with the chain number replaced by the total antenna count LM, written here as a network total,

$$\Xi_{\mathrm{pre}}^{\mathrm{cen}} = \frac{1}{\upsilon_{\mathrm{coh}}}\left(\frac{K^3}{3} + 3K^2LM + K\,LM\right). \tag{21}$$

Under local operation AP l instead inverts an $M\times M$ matrix, the same accounting with the antenna and user dimensions exchanged,

$$\Xi_{\mathrm{pre}}^{\mathrm{loc}} = \frac{1}{\upsilon_{\mathrm{coh}}}\left(\frac{M^3}{3} + 3M^2K + MK\right). \tag{22}$$

The splits distribute the two counts over the two nodes as

$$\Xi_i^{\mathrm{CPU}} = \iota_{\mathrm{S1}}\,2K\,LM + \iota_{\mathrm{S1,S2}}\,\iota_{\mathrm{DL}}\,\Xi_{\mathrm{pre}}^{\mathrm{cen}}, \tag{23}$$
$$\Xi_{i,l}^{\mathrm{AP}} = \iota_{\mathrm{S2,S3}}\,2K\,M + \iota_{\mathrm{S3}}\,\iota_{\mathrm{DL}}\,\Xi_{\mathrm{pre}}^{\mathrm{loc}}, \tag{24}$$

where $\iota_{\bullet}\in\{0,1\}$ equals one for the split or direction named in its subscript. The computation is charged to the downlink only, since TDD reciprocity lets the same matrix serve the uplink combiner, while the application is charged in both directions. Every operation appears exactly once across (23), so under S2 the AP is billed only for applying the precoder and the CPU only for computing it. The corresponding power is

$$P_{\mathrm{MIMO}} = \begin{cases} \bar{M} P_s + \dfrac{f_{s,\mathrm{I}}}{\eta}\,\Xi, & \Xi > 0, \\ 0, & \Xi = 0, \end{cases} \tag{25}$$

with η the computational efficiency of the precoder in the downlink and of the combiner in the uplink, P_s the static power of one FPGA and $\bar{M}$ the number of FPGAs, one per group of 32 chains: $\bar{M} = \lceil M/32 \rceil$ at an AP and $\bar{M} = \lceil LM/32 \rceil$ at the CPU, which processes the collective array. Charging no static power when $\Xi = 0$ matters for S1, where an AP has no precoder block at all and billing it for an idle FPGA would overstate precisely the split that is meant to be cheap on the AP side.

3) Central unit: Channel encoding and decoding stay at the CPU under every split, since that is where the payload enters and leaves the network, and are driven by the delivered network sum rates R_{DL} and R_{UL} through the encoder and decoder models P_{enc}, P_{dec} of [1]. With $\mathcal{A}_i$ the frame-averaging operator defined in (34), the central unit consumes

$$\begin{aligned} \overline{P_{\mathrm{CPU}}} = P_{\mathrm{CPU},0} + \mathcal{A}_{\mathrm{DL}}\big(P_{\mathrm{enc}} + P^{\mathrm{CPU}}_{\mathrm{MIMO,DL}}\big) \\ + \mathcal{A}_{\mathrm{UL}}\big(P_{\mathrm{dec}} + P^{\mathrm{CPU}}_{\mathrm{MIMO,UL}}\big), \end{aligned} \tag{26}$$

with $P_{\mathrm{CPU},0}$ the always-on consumption of the central unit.

B. Fronthaul and Synchronization

1) Fronthaul: Following the cell-free power model of [5], the link serving AP l consumes a traffic-independent part covering the optical transceivers at both ends and a part proportional to the rate it carries,

$$P_{\mathrm{FH},l} = P_{\mathrm{FH},0} + \Pi_{\mathrm{FH}}\, R_{\mathrm{FH},l}, \tag{27}$$

with Π_{FH} in W per bit/s. What $R_{\mathrm{FH},l}$ contains is set by the split of Section III-A1. Let b_{FH} be the number of bits per real sample, so a complex sample costs $2b_{\mathrm{FH}}$ bits. Under S1 the link carries M streams of frequency-domain samples at the constellation rate in both directions and in both the data and the signalling mode,

$$R^{\mathrm{S1}}_{\mathrm{FH},l} = 2\, b_{\mathrm{FH}}\, M\, f_{s,\mathrm{I}}. \tag{28}$$

Under S2 the downlink carries the delivered payload plus the precoding coefficients, an $M \times K$ complex matrix per subcarrier refreshed once per coherence block,

$$R^{\mathrm{S2,DL}}_{\mathrm{FH},l} = R_{\mathrm{DL}} + \frac{f_{s,\mathrm{I}}}{\upsilon_{\mathrm{coh}}}\, 2\, b_{\mathrm{FH}}\, M\, K, \tag{29}$$

where R_{DL} appears in full because every AP serves every user. Under S3 only the payload is sent over the fronthaullinks, $R^{\mathrm{S3,DL}}_{\mathrm{FH},l} = R_{\mathrm{DL}}$. In the uplink, linear combining is distributive over the APs, $\hat{s}_k = \sum_l \mathbf{v}^H_{kl}\mathbf{y}_l$, so an AP holding its combining coefficients forwards K scalar partial sums instead of M sample streams under both S2 and S3,

$$R^{\mathrm{S2,UL}}_{\mathrm{FH},l} = R^{\mathrm{S3,UL}}_{\mathrm{FH},l} = 2\, b_{\mathrm{FH}}\, K\, f_{s,\mathrm{I}}. \tag{30}$$

The signalling mode separates the centralized splits from the local one. Centralized operation needs the CPU to see the raw received pilots in order to estimate the collective channel, so during the uplink signalling phase the link carries $2b_{\mathrm{FH}} M f_{s,\mathrm{I}}$ under S1 and S2 alike, whereas under S3 the pilots are consumed locally and the signalling load vanishes. Centralization therefore imposes a sample-level fronthaul during pilot transmission independent of the data-phase split.

The link is bidirectional and stays up across the whole frame, so its static term is charged once per link with a single duty cycle rather than once per direction. With

$$\theta_l = \sum_{i \in \{\mathrm{DL,UL}\}} \tau_i \left[\tau_{i,\mathrm{sig}} + (1 - \tau_{i,\mathrm{sig}})\, \bar{x}_i\right] \tag{31}$$

the fraction of the frame in which the link is in a working mode, and $\delta^{\mu}_{\mathrm{FH}}$ the reduction factor of an idle transceiver,

$$\overline{P_{\mathrm{FH},l}} = \left[\theta_l + (1 - \theta_l)\, \delta^{\mu}_{\mathrm{FH}}\right] P_{\mathrm{FH},0} + \Pi_{\mathrm{FH}}\, \overline{R}_{\mathrm{FH},l}. \tag{32}$$

The averaged fronthaul rate reuses (34) with a zero base level, since an idle link carries no traffic,

$$\overline{R}_{\mathrm{FH},l} = R^{\mathrm{payload}}_{\mathrm{FH},l} + \sum_{i \in \{\mathrm{DL,UL}\}} \mathcal{A}_i\Big(R^{i}_{\mathrm{FH},l},\, R^{i,\mathrm{sig}}_{\mathrm{FH},l},\, 0\Big), \tag{33}$$

where $R^{\mathrm{payload}}_{\mathrm{FH},l}$ is the delivered downlink payload of the data-sharing splits S2 and S3 and is excluded from the averaging, the remaining arguments then being peak rates. The reason is that the map from rate to power in (27) is linear and $\mathcal{A}_i$ with a zero base is linear in its arguments, while the rate model already delivers R_{DL} with its prelog applied. Charging the delivered rate directly is thus equivalent to frame-averaging the corresponding peak rate, and doing both would count the frame structure twice. For the sample-forwarding split S1 the situation is the opposite: (28) is a hardware constant with no dependence on the delivered rate, and the frame structure is then the only mechanism that reduces the fronthaul load.

Remark 1 (Fronthaul ceiling on energy efficiency)**.** *Under a data-sharing split the same payload is duplicated to every AP, so the network fronthaul load is* $L\, R_{\mathrm{DL}}$. *Keeping only this term in* (19) *bounds the downlink energy efficiency by* $1/(\Pi_{\mathrm{FH}} L)$, *independently of the bandwidth, the transmit power and the precoder. With* $L = 16$ *and* $\Pi_{\mathrm{FH}} = 0.25\,\mathrm{W}$ *per Gbit/s the network cannot exceed* $250\,\mathrm{Mbit\,J^{-1}}$ *however well everything else is designed, which is why the serving cluster size is the dominant lever on cell-free energy efficiency.*

2) Synchronization: Coherent joint transmission across physically separate nodes requires the APs to share a frequency and phase reference and requires the TDD chains to be reciprocity-calibrated. Whether the reference is distributed over the fronthaul, over a dedicated timing network, or over the air, the AP must hold it whether or not the frame is carrying data. It is therefore considered as the constant P_{sync} of (20), entirely load-independent and outside the frame averaging, with its own supply-and-cooling efficiency $\eta_{\mathrm{sync,sc}}$.

What P_{sync} covers is the AP-side reference hardware only, that is the disciplined oscillator and the loop that slaves it to a reference. A shared master clock is not a per-AP cost and is counted in $P_{\mathrm{CPU},0}$ instead. The over-the-air part of reciprocity calibration is frame time and CPU computation rather than a continuous per-AP power, and is not modelled. TODO source a value for P_{sync} and $\eta_{\mathrm{sync,sc}}$. The plausible range for the reference hardware spans roughly two orders of magnitude, from a temperature-controlled oscillator slaved over the fronthaul to an oven-controlled one with its own GNSS receiver, and at $L = 32$ that range moves P_{net} by about 18 %. Report it as a sensitivity band rather than a point value. Over-the-air calibration also consumes frame time, which is not yet charged to $\tau_{i,\mathrm{sig}}$.

C. Access Point and Co-located Power Consumption

The three frame-averaged blocks of (20) come from the FR3 BS model of [1], evaluated once per AP. A distributed AP carries few antennas, so hybrid beamforming has nothing to amortise and the array is fully digital, $M_{\mathrm{RF}} = M_{\mathrm{ant}} = M$ with no phase shifters. Substituting this into the analog and amplifier assemblies of [1] reduces them term for term to the per-AP expressions below. The digital block cannot be reused directly from [1] because the encoder and decoder move to the CPU and the MIMO processing terms depend on the functional split.

1) Frame structure and operating-mode averaging: A TDD frame of duration T_f carries a downlink and an uplink phase with duration ratios $\tau_{\mathrm{DL}}, \tau_{\mathrm{UL}} \in [0, 1]$ and $\tau_{\mathrm{UL}} = 1 - \tau_{\mathrm{DL}}$. Within direction i the time splits into reference signalling, for a fraction $\tau_i \tau_{i,\mathrm{sig}}$, data transmission, for a fraction $\bar{x}_i \tau_i (1 - \tau_{i,\mathrm{sig}})$, micro-sleep, for $(1 - \bar{x}_i)\tau_i(1 - \tau_{i,\mathrm{sig}})$, and idle, for $(1 - \tau_i)$, with the resource load $\bar{x}_i$ of (1). The frame timing is shared by the whole network: under TDD the APs must switch direction together. In micro-sleep and idle a component drops to a fraction $\delta_c^{\mu}, \delta_c^{0} \in [0, 1]$ of a base level P^0. For a component c with working power P^{a} during data and P^{s} during signalling, the frame-averaged consumption of direction i is

$$\begin{aligned} \mathcal{A}_i^c(P^{\mathrm{a}}, P^{\mathrm{s}}, P^0) = {} & \bar{x}_i \, \tau_i \, (1 - \tau_{i,\mathrm{sig}}) \, P^{\mathrm{a}} + \tau_i \, \tau_{i,\mathrm{sig}} \, P^{\mathrm{s}} \\ & + (1 - \bar{x}_i) \, \tau_i \, (1 - \tau_{i,\mathrm{sig}}) \, \delta_c^{\mu} \, P^0 \\ & + (1 - \tau_i) \, \delta_c^{0} \, P^0. \end{aligned} \tag{34}$$

For the digital and analog blocks the three levels coincide, $P^{\mathrm{a}} = P^{\mathrm{s}} = P^0 = P$, and we write $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$; only the amplifier distinguishes them. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent and a load-independent part, which is what allows the consumption to be attributed to the delivered traffic.

TODO make this consistent so this can be removed The rate model states the frame as a coherence block of τ_c samples split into τ_p pilots, τ_u uplink data and τ_d downlink data, whereas the power model works in frame fractions. Charging the pilots to the uplink signalling phase pins the mapping,

$$\tau_{\mathrm{DL}} = \frac{\tau_d}{\tau_c}, \quad \tau_{\mathrm{DL,sig}} = 0, \quad \tau_{\mathrm{UL,sig}} = \frac{\tau_p}{\tau_p + \tau_u}, \tag{35}$$

so that $\tau_{\mathrm{UL}} \tau_{\mathrm{UL,sig}} = \tau_p / \tau_c$ is the pilot share and $\tau_{\mathrm{UL}}(1 - \tau_{\mathrm{UL,sig}}) = \tau_u / \tau_c$ the uplink data share. The same block length also fixes the precoder amortisation, $\upsilon_{\mathrm{coh}} = \tau_c$, so that a single number governs both the pilot overhead of the rate model and the reuse of the precoder in (21) and (22).

2) Power amplifiers: TODO check from here on out The amplifiers transmit only in the downlink. With output power p, the instantaneous consumption of one amplifier is

$$P_{\mathrm{PA}}(p) = \frac{1}{\eta_{\mathrm{PA,max}}} \left[\xi \, P_{\max} + (1 - \xi) \, P_{\max}^{1-\alpha} \, p^{\alpha} \right], \tag{36}$$

with $P_{\max}$ the maximum output power at a fixed back-off, $\eta_{\mathrm{PA,max}}$ the maximum efficiency, $\xi \in [0, 1]$ the static-to-dynamic ratio and $\alpha \in [0, 1]$ the efficiency exponent. Each AP is an independent transmitter with its own budget $\rho_{\max}$, so the single site power of [1] becomes the vector of radiated powers $P_{T,l}$ of (11), supplied by the rate model, which the two operation modes fill in differently: distributed operation meets the constraint with equality at every AP, $P_{T,l} = \rho_{\max}$, whereas centralized operation rescales all APs by one common factor and brings only the busiest of them to its budget. Assuming the AP power is shared equally across its antennas, $P_{a,l} = P_{T,l}/M$, and

$$\overline{P_{\mathrm{PA},l}} = M \, \mathcal{A}_{\mathrm{DL}}^{\mathrm{PA}}\big(P_{\mathrm{PA}}(P_{a,l}), \, P_{\mathrm{PA}}(\zeta_{\mathrm{sig}} P_{\max}), \, P_{\mathrm{PA}}(0)\big), \tag{37}$$

with ζ_{sig} the fraction of $P_{\max}$ used during signalling. The equal split is not exact, since the per-antenna powers differ for any non-isotropic precoder, but because $\alpha \in [0, 1]$ makes $P_{\mathrm{PA}}(\cdot)$ concave, Jensen's inequality gives $\frac{1}{M} \sum_m P_{\mathrm{PA}}(P_{a,lm}) \le P_{\mathrm{PA}}(P_{a,l})$: it is a conservative upper bound, tight when the precoder loads an AP uniformly.

Remark 2 (Amplifier sizing)**.** *The relation between $P_{\max}$ and the budget decides how the static term scales with the deployment and must be stated. We size each amplifier for its share of the budget it is actually given, $P_{\max} = \rho_{\max}/M$ at an AP and $P_{\max} = P_{\mathrm{TX}}/(LM)$ at the co-located site. Since the comparison holds the total transmit power fixed, $\rho_{\max} = P_{\mathrm{TX}}/L$, the two deployments then carry amplifiers of identical rating and differ only in how many sites they sit on. Substituting* (36) *into* (37) *makes the data-mode consumption of an AP independent of M,*

$$M \, P_{\mathrm{PA}}(P_{a,l}) = \frac{\rho_{\max}}{\eta_{\mathrm{PA,max}}} \left[\xi + (1 - \xi) \left(\frac{P_{T,l}}{\rho_{\max}} \right)^{\alpha} \right], \tag{38}$$

so the amplifiers see the precoders of Section II-C only through the fraction $P_{T,l}/\rho_{\max} \in [0, 1]$ of the budget that AP l radiates, and the network static floor is $L\xi\rho_{\max}/\eta_{\mathrm{PA,max}} = \xi P_{\mathrm{TX}}/\eta_{\mathrm{PA,max}}$, the same for both deployments at equal total transmit power. The per-AP constraint then costs the network twice under centralized operation: only part of the budget is *radiated, while the static floor is paid in full. TODO source $\xi, \alpha, \eta_{\mathrm{PA,max}}$ for AP-class amplifiers; the values used are fitted for macro-cell hardware.*

3) Analog front end: The analog block comprises, per RF chain, the digital-to-analog converters (DACs) in the downlink or analog-to-digital converters (ADCs) in the uplink, RF filters, mixers and the low-noise amplifiers, with the local oscillator shared across the chains of one node. Specialising [1] to a fully digital array without phase shifters,

$$P_{\mathrm{ana,DL},l} = P_{\mathrm{LO}} + M\left(2P_{\mathrm{DAC}} + 2P_{\mathrm{filt,RF}} + 2P_{\mathrm{mix}}\right), \quad (39)$$

$$P_{\mathrm{ana,UL},l} = P_{\mathrm{LO}} + M\big(2P_{\mathrm{ADC}} + 2P_{\mathrm{filt,RF}} + 2P_{\mathrm{mix}} + P_{\mathrm{LNA}}\big), \quad (40)$$

the factor two accounting for the in-phase and quadrature branches, and

$$\overline{P_{\mathrm{ana},l}} = \mathcal{A}_{\mathrm{DL}}^{\mathrm{ana}}(P_{\mathrm{ana,DL},l}) + \mathcal{A}_{\mathrm{UL}}^{\mathrm{ana}}(P_{\mathrm{ana,UL},l}) + P_{\mathrm{ana,misc}}. \quad (41)$$

This is the co-located analog block term for term, with no distributed addition: the synchronization sits outside it, in (20). The local oscillator is the first term to change character: a co-located array shares one across all M_{ant} chains, whereas the network pays $L\,P_{\mathrm{LO}}$.

4) Digital signal processing: Every AP performs the OFDM modulation and demodulation, the predistortion of its own amplifiers and the baseband filtering whatever the split, so with the component models P_{IFFT}, P_{DPD} and $P_{\mathrm{filt,BB}}$ of [1] evaluated at M chains,

$$P_{\mathrm{dig,DL},l} = P_{\mathrm{IFFT}} + P_{\mathrm{DPD}} + P_{\mathrm{filt,BB}} + P_{\mathrm{MIMO,DL},l}, \quad (42)$$

$$P_{\mathrm{dig,UL},l} = P_{\mathrm{IFFT}} + P_{\mathrm{filt,BB}} + P_{\mathrm{MIMO,UL},l}, \quad (43)$$

with $\overline{P_{\mathrm{dig},l}} = \mathcal{A}_{\mathrm{DL}}^{\mathrm{dig}}(P_{\mathrm{dig,DL},l}) + \mathcal{A}_{\mathrm{UL}}^{\mathrm{dig}}(P_{\mathrm{dig,UL},l})$. The encoder and decoder are absent by construction, which is the structural difference from the single-site model, and only the MIMO term of (25) depends on the split.

Remark 3 (Loss of static-power amortisation)**.** *The precoder, combiner and baseband filter share a static power P_s per FPGA, one per group of 32 chains. A co-located array of M_{ant} antennas amortises that power over 32 chains at a time, whereas a network of L APs with $M < 32$ antennas each pays it L times. At $L = 16$ and $M = 4$ the network runs 16 FPGAs where a co-located array of the same 64 antennas runs 2. Chopping a large array into many small ones leaves the per-chain digital costs untouched but destroys the sharing of the fixed ones, and this is a general feature rather than an artefact of the value 32.*

5) Co-located baseline: The reference deployment is the unmodified model of [1] evaluated at a single site: one fully digital array of $M_{\mathrm{ant}} = LM$ antennas radiating the full budget P_{TX}, one shared local oscillator, no fronthaul, no central unit and no synchronization power, with the complete digital chain (encoder, precoder, OFDM, predistortion and baseband filter in the downlink; decoder, combiner, OFDM and baseband filter in the uplink) at that one site. It is evaluated through the same parameter set as the distributed network, so the two deployments cannot silently differ in a hardware constant.

D. Delivered Rate and Energy Efficiency

The delivered sum rates R_{DL} and R_{UL} come from the rate model of Section II with the data-fraction prelogs of (18) already applied, and the network energy efficiency is the delivered rate per consumed watt,

$$\mathrm{EE}_{\mathrm{net}} = \frac{R_{\mathrm{DL}} + R_{\mathrm{UL}}}{P_{\mathrm{net}}} \quad [\mathrm{bit/J}], \quad (44)$$

which reduces to the single-site energy efficiency of [1] for the co-located baseline. Because a distributed and a co-located deployment cover the same area with the same hardware count, the comparison is made at equal LM and equal total transmit power.

TODO The encoder and decoder are driven by the *delivered* rates and their consumption is then frame-averaged again in (26), so the frame structure enters twice for those two components. This is inherited from [1] and is deliberately *not* done for the fronthaul payload in (33); decide which convention the paper adopts and apply it to both.

TODO Effects not yet captured. Fronthaul quantization is modelled only through its rate, so S1 is charged for its fronthaul load but not debited for the SINR loss the b_{FH} bits per sample cause. Channel estimation is unpriced, since perfect CSI is assumed. AP selection is not modelled: all L APs are always active, so the deep-sleep term of (19) and the serving-cluster reduction of Remark 1 are left out. Finally, the reduction factors, the supply-and-cooling efficiencies and the amplifier parameters are inherited from a macro BS and are the most likely source of systematic error when applied to a small outdoor AP.

IV. Numerical Evaluation

V. Conclusion

TODO future work:

- include channel estimation error in rate calculation
- include channel estimation power consumption in power model
- include pilot contamination in rate calculation
- make some parameters more realistic e.g. fronthaul power consumption, synchronization power consumption, etc.
- investigate the impact of hardware impairments on system performance
- accurately characterize all parameters based on the size of the BS/APs.

References

[1] E. Peschiera, S. Yun, Y. Lee, L. V. der Perre, and F. Rottenberg, “A parametric power model of upper mid-band (fr3) base stations for 6g,” in *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21 476–21 480.

[2] 3GPP, “Study on network energy savings for NR (release 18),” 3rd Generation Partnership Project (3GPP), Technical Report TR 38.864, Mar. 2023, version 18.1.0.

[3] ——, “Study on channel model for frequencies from 0.5 to 100 GHz,” 3rd Generation Partnership Project (3GPP), Technical Report TR 38.901, version 16.1.0, Dec. 2020. [Online]. Available: https://www.3gpp.org/DynaReport/38901.htm

[4] J. Hoydis, S. Cammerer, F. Ait Aoudia, A. Vem, N. Binder, G. Marcus, and A. Keller, “Sionna: An Open-Source Library for Next-Generation Physical Layer Research,” *arXiv preprint*, no. arXiv:2203.11854, 2022. [Online]. Available: https://arxiv.org/abs/2203.11854

[5] H. Q. Ngo, L.-N. Tran, T. Q. Duong, M. Matthaiou, and E. G. Larsson, “On the Total Energy Efficiency of Cell-Free Massive MIMO,” *IEEE Transactions on Green Communications and Networking*, vol. 2, no. 1, pp. 25–39, 2018.